

Threading

$P_1, P_2 \text{ fs} \Rightarrow$

program $\rightarrow$ executing state on the CPU $\rightarrow$ process

① Snake Game, vending m ②

2 process in T.M

SG $\rightarrow$ V.M { not directly }

SG $\rightarrow$ { file }, pipe { IPC expensive }

Uber $\rightarrow$ Google
APLS

Threads $\rightarrow$ V.M $\rightarrow$ P_a
 $\rightarrow$ t1 t2 t3

path of execution

V.M
 $\downarrow$
P1

SG
 $\downarrow$
P2

OS

CPU ✓ ✓
 $\uparrow$ $\uparrow$
t1 t2

fib(n)

10, 20, 30, ..., 100

FC

for () {
 fib(n)
}

✓ fib(10) ✓
✓ fib(20) ✓
✓ fib(30) - - -

main thread

main $\rightarrow$ t1 $\rightarrow$ t2 ✓ seq C1

$\xrightarrow{t1}$ C1 ✓
 $\xrightarrow{t2}$ C2 ✓
Parallelism

concurrent {
C1